

GAGAN THAKUR

Mandi, Himachal Pradesh

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EDUCATION

National Institute of Technology, Hamirpur

B.Tech in Electronics and Communication Engineering

Aug. 2024 – Aug. 2028 (Expected)

CGPA: 7.1 / 10

Swami Vivekanand Sr. Sec. School

High School

Apr. 2010 – Apr. 2023

Mandi, India

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python
Go Ecosystem: Gin, Gorilla WebSocket, net/http, Goroutines & Channels
Databases: MongoDB, PostgreSQL, SQL
Tools: Git, GitHub, Docker, GitHub Actions, Postman, AWS

EXPERIENCE

Open Source Contributor

Openchangelog (Go)

2024 – Present

Remote

- Debugged and improved a GitLab changelog source integration across the source, load, and parse pipeline.
- Collaborated with maintainers through GitHub discussions, code reviews, and issue tracking.

PROJECTS

Kairo – Collaborative Text Editor Plugin [\[GitHub\]](#)

Go · Gorilla WebSocket · Operational Transformation

- Convert any normal Text Editor into Collaborative .
- Built a WebSocket server in Go that keeps multiple clients in sync on a shared document using Operational Transformation, reconciling concurrent edits by version.
- Implemented live cursor tracking and session management, with sessions created on-demand and identified by a unique ID.

Go-edis – Redis Server Built from Scratch [\[GitHub\]](#)

Go · TCP · RESP Protocol

- Implemented a Redis-compatible server from scratch in Go, including a raw TCP listener and a parser for the RESP protocol.
- Built an in-memory key-value store handling concurrent client connections, testable directly with `redis-cli`.

Restaurant-Backend – REST API for Restaurant Management [\[GitHub\]](#)

Go · Gin · MongoDB · JWT

- Designed a REST API in Go/Gin for managing users, tables, menus, foods, orders, and invoices, backed by MongoDB.
- Implemented JWT-based authentication with access and refresh tokens, and protected routes for restaurant operations.

Synomia – Persistent Free Communication [\[GitHub\]](#) | [\[Live\]](#)

Go · Gorilla WebSocket · Next.js (Frontend)

- Built a WebSocket-based chat server in Go with a hub/session architecture supporting multiple concurrent chat rooms.
- Handled client registration, broadcast messaging, and typing indicators via Go channels, paired with a Next.js frontend.

ACHIEVEMENTS

- Solved **250+ Data Structures and Algorithms** questions and completed the **NeetCode 150** roadmap twice.
- Participated in multiple hackathons including Smart India Hackathon and ComputeHive.

INTERESTS

- Reverse Engineering
- Computer Science across different fields
- Art
- Literature